Web GIS

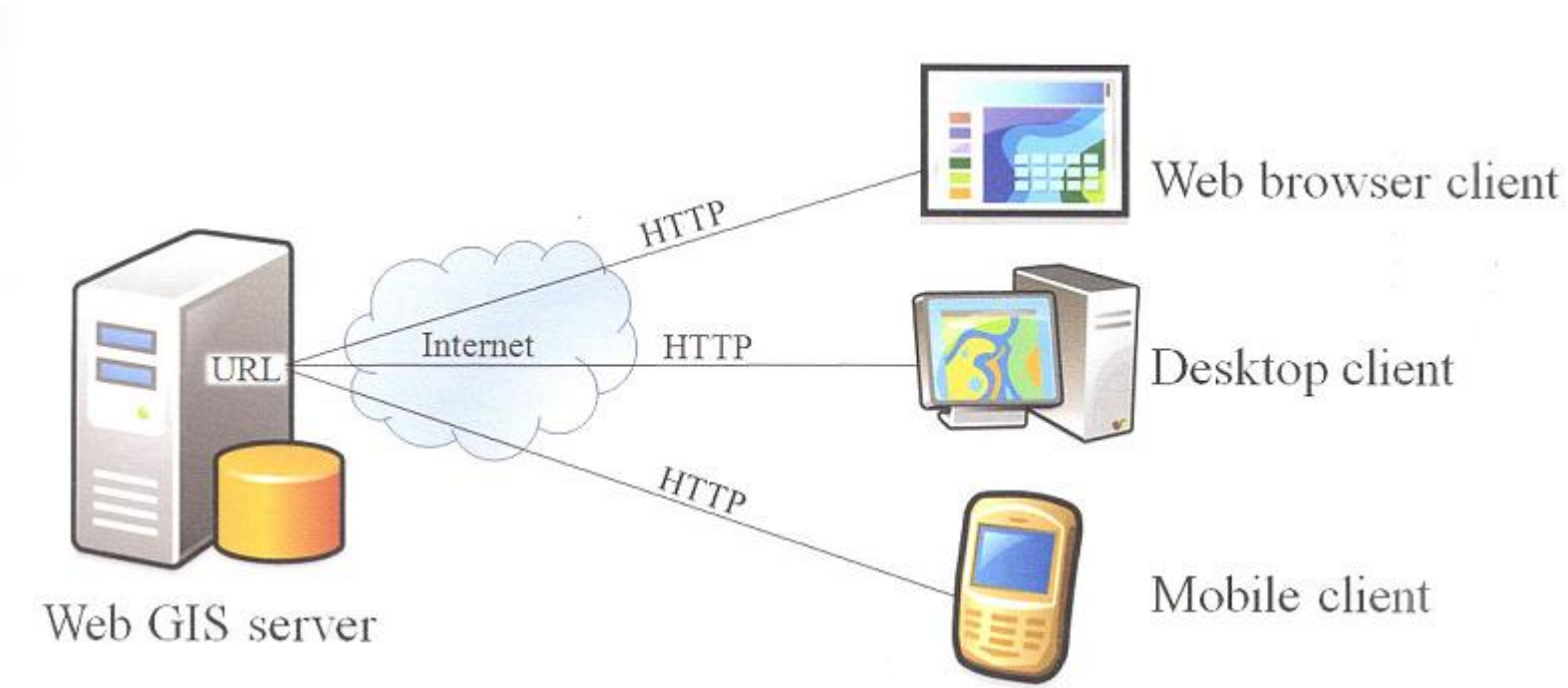
Lecture 2

Web GIS

- Definition
  - Web GIS is a type of distributed information system.
  - The simplest form of Web GIS should have at least a server and a client
  - Web Cycle

Web GIS

- Web GIS is any GIS that uses Web technology to communicate between components

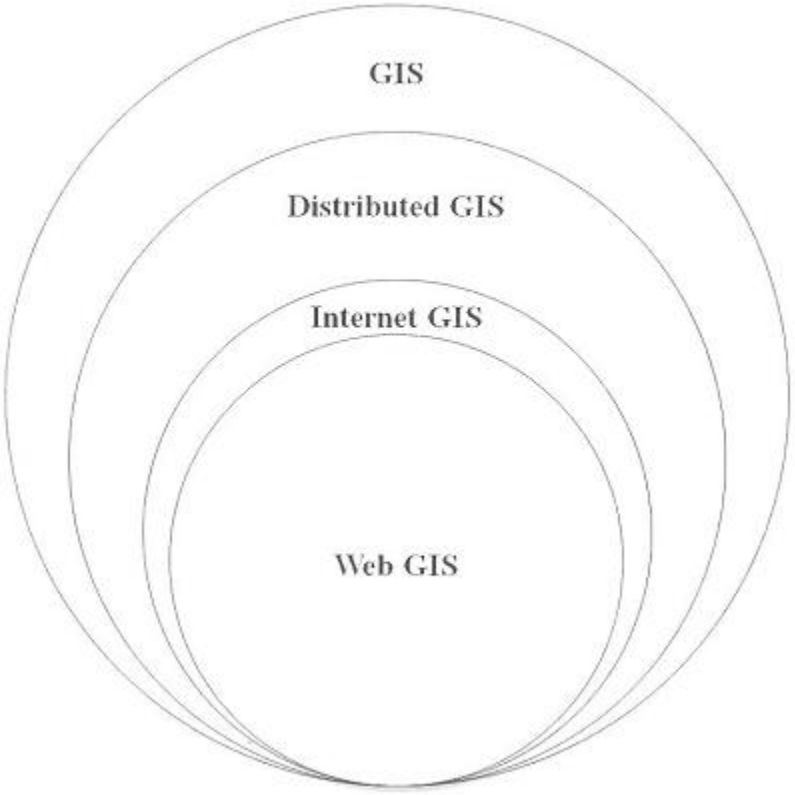


Web GIS

Web GIS is further defined by

- HTTP – is the main protocol used by the different components of Web GIS to communicate with each other.
- The two tier system that involves a server and one or more clients
- Many Web GIS consists of three tiers, that include a data tier
- Web GIS and desktop GIS are increasingly intertwined. Web GIS relies on desktop GIS to author resources.

Web GIS



Web GIS

- Internet GIS vs Web GIS
  - Internet GIS supports many services, and Web GIS is one of them

Web GIS

- Advantages of Web GIS over traditional desktop GIS
  - A global reach
  - A large number of users
  - Better cross-platform capability
  - Low cost as averaged by the number of users
  - Easy to use for end users
  - Unified update
  - Diverse applications

Web GIS

- Web GIS is able to perform all GIS functions involving spatial information, including **capture**, **storage**, **editing**, **manipulation**, **management**, **analysis**, **sharing** and **visualization**

Web GIS

- Strengths of GIS
  - Mapping (visualization) and query
  - Collection of geospatial information
    - VGI (Goodchild, 2007)
  - Dissemination of geospatial information
  - Geospatial analysis

Web GIS

- Uses
  - Web GIS as a new business model and new type of commodity
  - Web GIS as an engaging and powerful tool for e-government (USGS)
  - A new infrastructure for e-science

Web GIS

- Reading assignment
  - 1.4.3 Web GIS Opportunities